MIL-B-43611A

30 September 1974

SUPERSEDING
MIL-B-43611
29 November 1968

MILITARY SPECIFICATION

BAG, PAPER: WAXED (FIELD BREAD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers waxed paper bags (see 6.1).

* 1.2 Classification. Waxed paper bags covered by this specification shall be of the following types and minimum sizes as specified (see 6.2 and 6.3).

Type I - Self opening (automatic), 13 by 6 by 33 inches (Face, width, height).

Type II - Square, 13 by 6 by 35 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-B-36 - Bags, Paper (Grocers).
UU-P-268 - Paper, Kraft, Untreated, Wrapping.
* PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 8105

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(Copies of specifications, standards and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* LAWS AND REGULATIONS

21 CFR 121 - Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical Association of the Pulp and Paper Industry (TAPPI)

T-401 - Fiber Analysis of Paper and Paperboard
T-402 - Conditioning Paper and Paperboard for Testing
T-403 - Bursting Strength of Paper
T-404 - Tensile Breaking Strength of Paper and Paperboard

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T-405 - Wax in Papers
T-410 - Basis Weight of Paper and Paperboard
T-414 - Internal Tearing Resistance of Paper
T-477 - Blocking Resistance of Paper and Flexible Materials
T-483 - Odor of Packaging Materials
T-630 - Melting Point (Plateau) of Petroleum Wax

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30346.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. The materials used in the fabrication of waxed paper bags shall conform with the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

* 3.1.1 Waxed kraft paper. The paper used in the production of waxed paper bags shall be not less than 90 percent unbleached kraft fibers, and not more than 10 percent of other chemical fibers providing the paper meets the physical requirements of table I. The paper shall be waxed both sides. Physical requirement tests shall be as specified in 4.4.1 and table VII.

* TABLE I. Minimum physical requirements of waxed kraft paper

Characteristics	Requirements - minimum
Wax in paper	12 pounds per ream 1/
Basis weight of unwaxed paper	50 pounds per ream 1/, minus 5 percent, unlimited plus tolerance
Bursting strength (before waxing)	31 points per square inch
Tensile strength	17 pounds per inch of width
Tearing resistance, machine direction (before waxing)	92 grams
Blocking resistance	Slight blocking
Odor of packaging material	No evidence of foreign odor
Melting point of wax	140°F.

1/ Ream (24 by 36 inches - 500 sheets).

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3.1.2 Adhesive. The adhesive used for the seams of waxed paper bags shall be waterproof to the extent that there shall be not more than 25 percent adhesive failure of the bonded area when tested as specified in 4.4.2. Seams shall be continuously bonded.

* 3.2 Construction. Waxed paper bags shall be constructed with an open mouth in accordance with applicable figures in UU-B-36. The bags shall be fabricated with a thumb notch or lip in accordance with commercial practice.

* 3.3 Dimensions. Dimensions specified in 1.2 shall be minimum sack dimensions when measured as specified in applicable figures in UU-B-36. The edges of unopened bags shall be uniform and parallel.

* 3.4 Marking. Each bag shall be printed or stenciled across one face of the bag with letters not less than 1/2 inch high with the words "FIELD BREAD BAG". The ink used in printing or stenciling shall be commercially waterfast and sunfast and shall have no objectionable odor. The position of the printing shall be not less than 10 inches from the open top of the bag and approximately equidistant from each edge of the bag.

3.5 Workmanship. Bags shall be uniformly constructed and shall be free from holes, tears, or other defects which would impair their usefulness. Bags shall conform to the levels of quality established herein.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

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* 4.2.1.1 Testing of components and materials. Testing of materials described in this specification shall be as indicated in table II. The lot for the purpose of determining sample size shall be expressed in units of rolls of the waxed kraft paper. The sample unit shall be one piece the full width of roll by 36 inches long. Not more than one sample unit shall be drawn from any one roll and shall not be taken from the two outer coils of the roll. The inspection level shall be S-1, except that minimum sample size shall be three sample units. Results for the characteristics shown in table II shall apply to lot average. The lot shall be subject to rejection if any of the average minimum requirements of the waxed kraft paper are not met.

4.2.1.2 Conditioning of the test unit. Before testing for the characteristics in table II, all test units shall be conditioned to equilibrium in accordance with TAPPI Method T-402. Conditioning shall also apply on obtaining the basis weight of kraft paper after the wax extraction cycle.

TABLE II. Instructions for testing component materials

Characteristic	Reqmt. of paragraph	Test method	Determinations per sample unit	Report to nearest
Fiber analysis of kraft paper	3.1.1	4.4.1		1.0 percent
Weight of wax in paper	3.1.1	4.4.1	2	0.1 pound
Basis weight of paper after wax extraction	3.1.1	4.4.1	2	0.1 pound
Bursting strength	3.1.1	4.4.1	Average of 10-5 on each side	0.1 point
Tensile strength	3.1.1	4.4.1	Average of 5	0.1 gram
Tearing resistance (machine direction)	3.1.1	4.4.1	Average of 5	1.0 gram
Blocking resistance	3.1.1	4.4.1	1	rating
Melting point of wax	3.1.1	4.4.1	1	1°F.
Odor (foreign)	3.1.1	4.4.1	1	none

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4.3 Inspection of the end item.

* 4.3.1 Examination of the end item. Examinations shall be made for defects in table IV and 4.3.1.1. The sampling plans and acceptable criteria shall be as shown in table III. The acceptable quality levels (AQLs) shall be expressed in terms of defects per 100 units.

* TABLE III. Sampling plans and acceptable quality levels

Examination in	Inspection level	Sample unit	Lot size expressed in	AQLs
Table IV	S-4	One bag	Bags	6.5
4.3.1.1	S-2	One bag	Bags	4.0

TABLE IV. End item visual defects

Examine	Defect
Construction	Bag not type specified (see 1.2). Seams not continuously bonded (see 3.1.2). Edges of unopened bag not uniform and parallel (see 3.3). Thumb notch or lip missing (see 3.2).
Marking	Not as specified (see 3.4). Letters less than 1/2 inch high (see 3.4). Not legible (see 3.4). Position of printing less than 10 inches from top of bag, not equidistant from bag edges.
Workmanship	Tears or holes (see 3.5). Surface not clean (see 3.5).

4.3.1.1 Dimensional examination. Any dimension of the bag that is less than the specified size (see 3.3) shall be classified as a defect.

* 4.3.1.2 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 are complied with. Defects shall be as specified in table V. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

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TABLE V. Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Material	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as; incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted containers. Inadequate securing or sealing of wrapping materials and containers.
Content	Number of bags per shipping container is more or less than required. Bags of different type in same package. Number of bags per package not as required.

4.3.2 Testing of the end item. The end item shall be tested for the characteristic listed in table VI for each lot presented for examination at one time. The sample unit shall be one bag. The sample size shall be as indicated by inspection level S-1 and the acceptable quality level (AQL) shall be 6.5 expressed as percent defective. Tests results are applicable to average unit requirements.

TABLE VI. Instructions for testing end item

Characteristic	Reqmt. of paragraph	Test method	Determinations per sample unit	Results reported as pass or fail 2/
Waterproofness of adhesive 1/	3.1.2	4.4.2	2 (one on side and bottom seam specimen)	X

1/ Failure of any specimen to pass the test shall be cause for rejection of the lot.

2/ If failure is indicated, report description thereof.

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4.4 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.4.1 Component testing (see table I). After conditioning the specimen as specified in 4.2.1.2, the specimens shall be tested in accordance with the TAPPI methods in table VII.

TABLE VII. Test methods of TAPPI

Test	Method
Fiber Analysis of Paper and Paperboard	T-401
Wax in paper	T-405
Basis weight of unwaxed paper	T-410
Bursting strength	T-403
Tearing resistance	T-414
Tensile breaking strength	T-404
Blocking resistance	T-477
Odor	T-483
Melting point of wax	T-630

4.4.2 Testing of the end item for water resistance of adhesive (see 3.1.2). Cut strips one-inch wide across the longitudinal and bottom seams, adjusting the length so that both sheetings will extend approximately 1-1/2 inches beyond each edge of the bonded joint. Immerse the specimens in approximately 1-1/2 inches of distilled water at room temperature for not less than 24 hours; after the bath cycle, remove one specimen, blot off excess water with blotting paper, and test before removing the next strip. To test, grasp the projecting ends of the specimen with one thickness of paper between the thumb and forfinger of each hand, the forefingers resting against

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the bonded joint, one on each side. Separate the joints by gently pulling the ends apart with a rolling motion from the top downward with second joints of the second and third fingers successively resting against each other to act as fulcrums. Examine the manner of separation of the joint, noting the percent of separation that occurs without fiber failure of the prebonded area of each separated sheet. Failure of more than ten percent of the specimens tested shall be cause for lot rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Two hundred and fifty bags of one type only shall be uniformly stacked and assembled into a package. The package shall then be completely wrapped in the same material as specified for the bag (see 3.1.1) or in kraft paper having a minimum basis weight of 60 lbs. per ream (24 by 36 inches-500 sheets) in accordance with grade B of UU-P-268. The paper wrap shall overlap itself widthwise and lengthwise a minimum of four inches and shall be secured with crossties of twine. The twine shall have a minimum breaking strength of 200 lbs. and the two intersections of the cord shall be interlocked. Alternatively, the bags may be stacked in quantities of 50 and either crosstied with not less than 50 pound breaking strength cord or twine or secured with a kraft paper band or sleeve having a minimum basis weight of 30 pounds per ream (25 by 36 inches-500 sheets). Two hundred and fifty bags shall then be packaged in a snug-fitting bag made from the material specified for the 250 bag wrap above. Closure of the bag shall be effected by double folding over the mouth a minimum of 1/2 inch and either gluing, stapling with not less than three staples, or taping with tape conforming to PPP-T-76.

* 5.1.2 Level C. Bags of one type only shall be packaged to afford adequate protection against physical damage during shipment from the supplier to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the supplier for retail distribution.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Five hundred bags of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, end opening, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with PPP-B-636.

* 5.2.2 Level B. Five hundred bags of one type only, packaged as specified in 5.1, shall be baled by wrapping with two sheets of kraft paper each sheet having a basis weight of not less than 60 pounds per ream in accordance with

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grade B of UU-P-268. The wraps shall overlap widthwise and lengthwise a minimum of four inches and shall be secured with tape conforming to PPP-T-76. Alternatively, five hundred bags may be packed directly into a fiberboard shipping container conforming to style RSC, end opening, type CF, variety SW, or type SF, class domestic, grade 350 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fabricated and closed in accordance with PPP-B-636.

* 5.2.3 Level C. Bags packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the supplier for retail distribution. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special markings required in the contract or order, shipping containers and bales shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Waxed paper bags are intended for use in the delivery of 6-loaf sheets of field bread from the mobile or field bakery to the using units.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of bag (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1).
(d) When weather-resistant grade fiberboard shipping containers are required for level B shipments (see 5.2.2).

* 6.3 The type III bag has been deleted because of no known users requirements of FSNs.

* 6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evalute the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Bag, Paper; Waxed (Field Bread) MIL-B-43611A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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